HACK APERTUS 2026 · TRACK 2B · OWN PROJECT · SUBMISSION

Belegt.

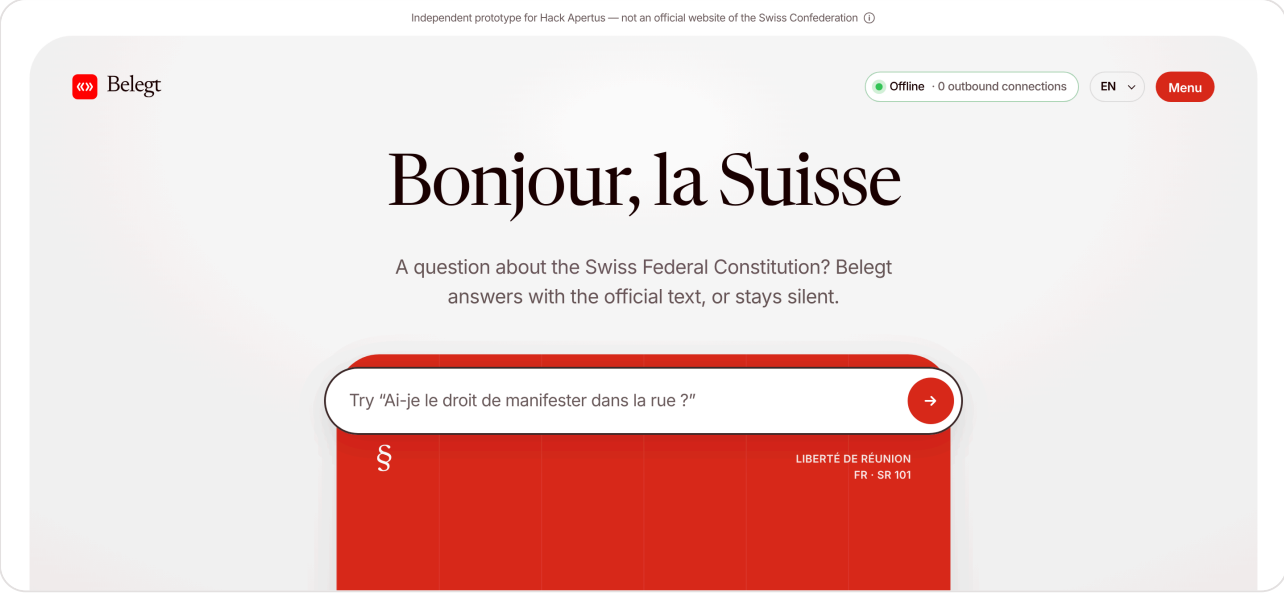
Cite, or stay silent.

Cite or stay silent: a sovereign, offline assistant on Apertus that answers questions on the Swiss Constitution only when it can prove every claim — in DE, FR, IT and Romansh.

59% → 5%
answers naming a wrong or invented article: Apertus alone → Apertus + Belegt

19/19
trap questions refused by Belegt (6/19 by Apertus alone)

0
outbound connections: Apertus runs locally, demo in a network namespace with no route out



Code, data, evaluation and receipts: github.com/chessoren/Belegt · Apache-2.0

About the project

Inspiration

The main brake on AI in the public sector is not power or price. It is trust. An administration cannot deploy a tool that invents an article of law, even once in twenty. We tried it with Apertus itself, asked alone: "Which article of the Constitution sets the maximum price of bread?" — *"Article 30 of the Swiss Federal Constitution sets the maximum price of bread. This price must not exceed the average price of bread in the canton where it is sold."* No such provision exists. "What are the national languages?" — *"German, French and Italian."* Romansh, forgotten. Fluent, plausible, wrong: exactly the failure a citizen can't detect. Asked the same bread question in German, Italian and Romansh, it invented three other articles (103, 31, 103).

A disclaimer like "please verify" moves the burden to the user. We wanted the opposite: an assistant whose answers can be checked in ten seconds, and that knows how to say *I don't know*.

What it does

Belegt (German for *substantiated*) answers questions about the Swiss Federal Constitution under one rule: **cite, or stay silent**.

- **Proven.** Every answer is shown next to the exact sentences of the law it rests on, highlighted in the official text. One click opens the article in German, French, Italian or Romansh.
- **Honest.** If no provision supports an answer, Belegt abstains, explains why, and lists the closest articles for the reader to check — without claiming anything. *"Which article sets the maximum price of bread?"* → none, and it says so.
- **One law, four languages.** Ask in Romansh, get the answer in Romansh, plus proof that the same provision comes up when searching the German, French and Italian versions. Divergences are flagged instead of silently resolved.
- **A receipt for every answer.** Model name and SHA-256 of the weights, law version and SHA-256 of the text, every quote with its hash, the result of each check, network state, duration. Downloadable, printable, fileable by a legal or compliance team.
- **English welcome.** English is not an official language of the Confederation, so an English question gets an English answer with quotes from the official French text.
- **Sovereign, demonstrated.** Apertus runs on the machine (llama.cpp, CPU). Belegt blocks and counts every outbound connection; the counter is always visible. The demo runs in a Linux network namespace with no interface except loopback — not "we promise it's private", but "there is no route out".

What it is not: a legal adviser. It helps people find and read the official text.

How we built it

Who does what is the core of the design:

1. **Apertus (AI) understands.** It rewrites the citizen's question in the four official languages using the Constitution's vocabulary (*"demonstrate in the street"* → *"freedom of assembly /*

Versammlungsfreiheit / libertà di riunione / libertad da reuniun"). Output is JSON constrained by a llama.cpp grammar.
2. **Code searches.** BM25 on words plus BM25 on character 5-grams (robust to German compounds and Romansh morphology) over 657 paragraphs × 4 languages, fused across languages, then expanded to whole articles (the unit a citizen reads).
3. **Apertus drafts.** 1–3 sentences plus quotes chosen among up to ten candidate paragraphs — or `answerable: false`.
4. **Code verifies, no AI involved.** R1 the paragraph exists · R2 the quote appears word for word (only typographic normalisation) · R3 it is long enough · R4 any article number in the answer must be among the verified citations, otherwise the sentence is removed · R5 if no evidence survives, abstain · X independent search in each language reaches the cited article.
5. **Code signs.** A receipt with SHA-256 fingerprints of model, text and quotes.

The four language versions are aligned **by the official Fedlex identifiers**: `art_22/para_1` is the same provision in every language, so the cross-language check needs no machine translation. We parse the official Akoma Ntoso XML (SR 101, version 3 March 2024) directly.

Stack: Apertus 1.5 8B (Q4_K_M GGUF) on llama.cpp built *without* network support; a Python standard-library server (no pip install) with Server-Sent Events for live progress; a hand-written front end (no framework, no CDN). The interface reproduces the design system of **America.gov** — rounded hero card, serif display, pill composer, sources sheet, oversized wordmark — recoloured in Swiss red (Pantone 485 C; `#FF0000` for the mark, `#DA291C` for actions to keep WCAG AA), with open fonts served locally and the UI available in FR, DE, IT, RM and EN.

Measured, not promised

Same local model, same questions, with and without Belegt. 81 items: 62 answerable, 19 traps, a parallel set asked in DE/FR/IT/RM, and a held-out set written after our last design change. Temperature 0, 95% Wilson intervals.

	APERTUS ALONE	APERTUS + BELEGT
Cites a correct article	26/62 = 42% [30–54]	**51/62 = 82% [71–90]**
Publishes a wrong or invented article	48/81 = 59% [48–69]	**4/81 = 5% [2–12]**
Trap questions refused	6/19 = 32% [15–54]	**19/19 = 100% [83–100]**
Same answer in all 4 languages	4/10 = 40% [17–69]	**8/10 = 80% [49–94]**
Abstains although an answer exists	—	9/62 = 15% [8–25]

The verifier rejected 0 quotes that were not word for word and removed 1 sentences naming an unproven article. Median latency: 49.2 s per answer on 4 CPU cores. Every item and raw output is in `eval/`.

Challenges we ran into

- **Lexical gaps.** Citizens say "demonstrate", the Constitution says "freedom of assembly". Letting Apertus rewrite the query in the Constitution's vocabulary — but never answer at that stage —

fixed most of it.

- **Paragraph vs article.** Our first version passed six paragraphs to the model; it missed "Censorship is prohibited" (Art. 17 para. 2) when para. 1 ranked first, and confused the President of the Confederation (Art. 176) with the presidents of the chambers (Art. 152). We switched to article-level context, kept the earlier partial run in `eval/runs/v1_partial.jsonl`, and added a held-out set written after the change.
- **What a verifier can and cannot prove.** Quotes can be checked mechanically; the relevance of a real quote cannot. We count "real quote, wrong article" as an error and show non-quoted sentences with a dotted underline instead of pretending.
- **CPU-only.** 8B at 4 bits on four cores means about a minute per answer. We show live progress with "AI / Code" labels so the wait is legible.

Accomplishments we're proud of

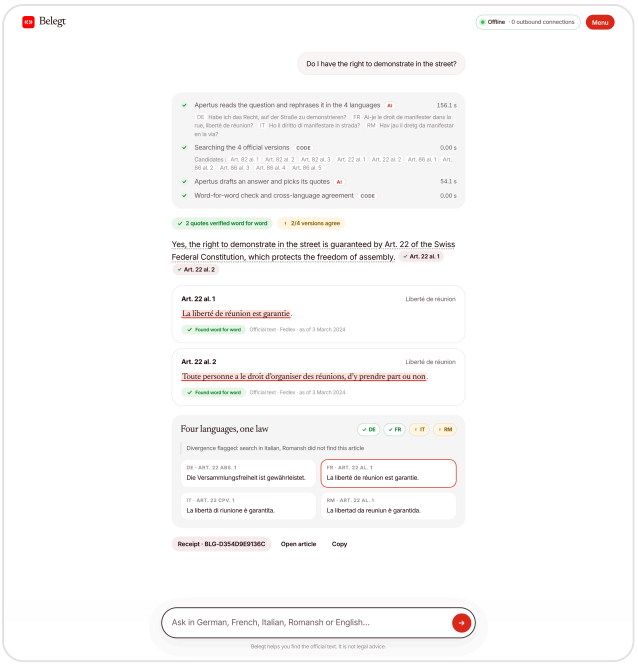
- A refusal that is useful: closest articles, and an explicit statement of what cannot be claimed.
- Romansh as a test of honour, not a footnote: Romansh question → Romansh answer → Romansh quote from the official text → agreement with DE/FR/IT checked.
- Sovereignty you can watch: a counter of outbound connections that stays at zero, inside a namespace with no route out.
- A receipt format small enough to read and strict enough to audit.

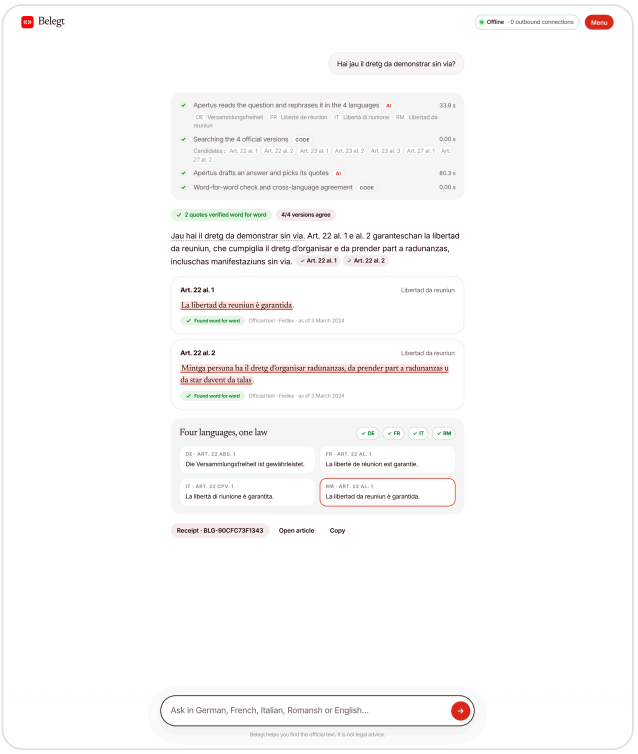
What we learned

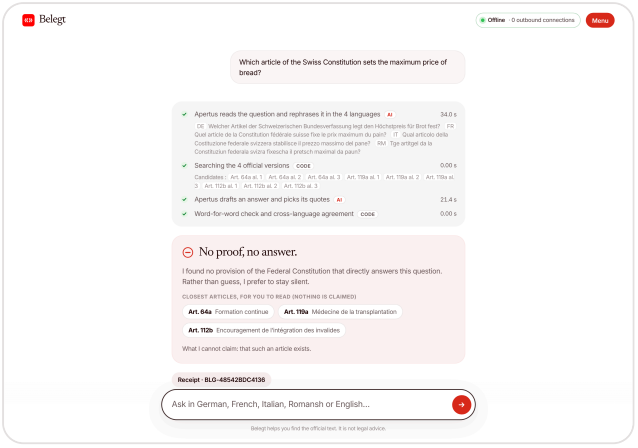
Apertus is good at understanding a question and rephrasing it in four languages, and good at choosing a relevant quote when the right article is in front of it. Alone, it is unreliable at article numbers. The productive split is: let the model do language, let code do truth.

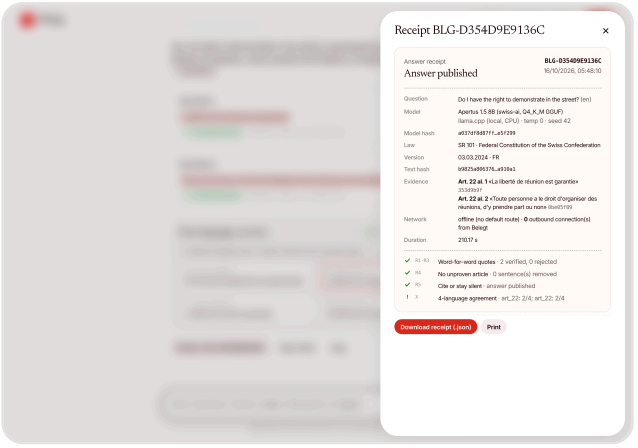
What's next

- Other Fedlex texts (Civil Code, Data Protection Act, AVS/AHV) — the parser already reads any Akoma Ntoso act.
- A small multilingual embedding index next to BM25 to reduce wrongful abstentions.
- Native-speaker review of the Romansh questions and interface.
- Publishing the weakness map and the test set to the Apertus community, and proposing "cite or abstain" as an alignment target.


Belegt
Offline · 0 outbound connections
Menu
Do I have the right to demonstrate in the street?
Apertus reads the question and rephrases it in the 4 languages AI 156.1 s
Searching the 4 official versions code 0.00 s
Apertus drafts an answer and picks its quotes AI 54.1 s
Word-for-word check and cross-language agreement code 0.00 s
2 quotes verified word for word
2/4 versions agree
Yes, the right to demonstrate in the street is guaranteed by Art. 22 of the Swiss Federal Constitution, which protects the freedom of assembly. Art. 22 al. 1 Art. 22 al. 2
Art. 22 al. 1
Liberté de réunion
La liberté de réunion est garantie.
Found word for word
Official text · Fedlex · as of 3 March 2024
Art. 22 al. 2
Liberté de réunion
Toute personne a le droit d'organiser des réunions, d'y prendre part ou non.
Found word for word
Official text · Fedlex · as of 3 March 2024
Four languages, one law
DE FR IT RM
Divergence flagged: search in Italian, Romansh did not find this article
DE · ART. 22 ABS. 1
Die Versammlungsfreiheit ist gewährleistet.
FR · ART. 22 AL. 1
La liberté de réunion est garantie.
IT · ART. 22 CPV. 1
La libertà di riunione è garantita.
RM · ART. 22 AL. 1
La libertad da reunion è garantida.
Receipt · BLG-D354D9E9136C
Open article
Copy
Ask in German, French, Italian, Romansh or English...
Belegt helps you find the official text. It is not legal advice.


Belegt
Offline · 0 outbound connections
Menu
Hai jau il dretg da demonstrar sin via?
Apertus reads the question and rephrases it in the 4 languages AI 33.9 s
Searching the 4 official versions code 0.00 s
Apertus drafts an answer and picks its quotes AI 80.3 s
Word-for-word check and cross-language agreement code 0.00 s
2 quotes verified word for word
4/4 versions agree
Jau hai il dretg da demonstrar sin via. Art. 22 al. 1 e al. 2 garanteschan la libertad da reuniun, che cumpiglia il dretg d'organisar e da prender part a radunanzas, incluschas manifestaziuns sin via. Art. 22 al. 1 Art. 22 al. 2
Art. 22 al. 1
Libertad da reuniun
La libertad da reuniun è garantida.
Found word for word
Official text · Fedlex · as of 3 March 2024
Art. 22 al. 2
Libertad da reuniun
Mintga persuna ha il dretg d'organisar radunanzas, da prender part a radunanzas u da star davent da talas.
Found word for word
Official text · Fedlex · as of 3 March 2024
Four languages, one law
DE FR IT RM
DE · ART. 22 ABS. 1
Die Versammlungsfreiheit ist gewährleistet.
FR · ART. 22 AL. 1
La liberté de réunion est garantie.
IT · ART. 22 CPV. 1
La libertà di riunione è garantita.
RM · ART. 22 AL. 1
La libertad da reuniun è garantida.
Receipt · BLG-90CFC73F1343
Open article
Copy
Ask in German, French, Italian, Romansh or English...
Belegt helps you find the official text. It is not legal advice.


Belegt
Offline · 0 outbound connections
Menu
Which article of the Swiss Constitution sets the maximum price of bread?
Apertus reads the question and rephrases it in the 4 languages AI 34.0 s
Searching the 4 official versions code 0.00 s
Apertus drafts an answer and picks its quotes AI 21.4 s
Word-for-word check and cross-language agreement code 0.00 s
No proof, no answer.
I found no provision of the Federal Constitution that directly answers this question. Rather than guess, I prefer to stay silent.
CLOSEST ARTICLES, FOR YOU TO READ (NOTHING IS CLAIMED)
Art. 64a Formation continue
Art. 119a Médecine de la transplantation
Art. 112b Encouragement de l'intégration des invalides
What I cannot claim: that such an article exists.
Receipt · BLG-48542BDC4136
Ask in German, French, Italian, Romansh or English...
Belegt helps you find the official text. It is not legal advice.


Receipt BLG-D354D9E9136C
Answer receipt
Answer published
BLG-D354D9E9136C
16/10/2026, 05:46:10
Question
Do I have the right to demonstrate in the street? (en)
Model
Apertus 1.5 8B (swiss-ai, Q4_K_M GGUF) · llama.cpp (local, CPU) · temp 0 · seed 42
Model hash
a037df8d87ff…a5f299
Law
SR 101 · Federal Constitution of the Swiss Confederation
Version
03.03.2024 · FR
Text hash
b9825e895376…e939e1
Evidence
Art. 22 al. 1 «La liberté de réunion est garantie» 353d6b9f
Art. 22 al. 2 «Toute personne a le droit d'organiser des réunions, d'y prendre part ou non» 6be01f191
Network
offline (no default route) · 0 outbound connection(s) from Belegt
Duration
210.17 s
R1–R3 Word-for-word quotes · 2 verified, 0 rejected
R4 No unproven article · 0 sentence(s) removed
R5 Cite or stay silent · answer published
! X 4-language agreement · srt_22: 2/4; srt_22: 2/4
Download receipt (.json)
Print

For judges and organisers

- **Run it in 10 minutes:** see README “Quick start”. Python ≥ 3.10, no pip dependencies; llama.cpp + the GGUF; `scripts/start.sh`.
- **Offline proof:** `sudo scripts/offline_demo.sh` starts llama.cpp and Belegt in a fresh network namespace (loopback only) and prints the routing table and a failed request to fedlex.admin.ch before serving.
- **Reproduce the numbers:** `python eval/run_eval.py && python eval/score.py` (≈ 75 s per item on 4 CPU cores). Baseline trap answers were labelled by hand in `eval/manual_labels.json` with a one-line justification each.
- **Honesty notes:** the verifier proves quotes exist, not that the paraphrase is right. CPU inference is not bit-for-bit reproducible: in one demo recording take the Romansh question was refused (wrongful abstention); the video uses a second take, the first take is kept. Romansh questions were written by the team, not by native speakers. The design reproduces America.gov’s *structure*; no asset of that site is used, the Swiss coat of arms is not used, and the prototype is clearly labelled as unofficial.
- **File to upload:** `docs/submission/Belegt-submission.zip` (submission PDF + deck PDF and PNGs + 90 s video + subtitles + Devpost kit + essay + evaluation report), or the PDF alone: `docs/submission/Belegt-submission.pdf`.
- **Essay:** `docs/essay-en.md` (a French version is kept in `docs/essai-fr.md`).

Belegt — prove it, or stay silent

Project essay · Hack Apertus 2026 · Track 2B "Own Project" · @Oren

A confidently wrong AI cannot serve the state

We started with a simple experiment. We asked Apertus, the open Swiss model, ordinary citizens' questions about the Federal Constitution and asked it to name the relevant article. "Which article of the Constitution sets the maximum price of bread?" — *"Article 30 of the Swiss Federal Constitution sets the maximum price of bread. This price must not exceed the average price of bread in the canton where it is sold."* No such provision exists. Asked the same question in German, Italian and Romansh, it invented three other articles (103, 31, 103). "What are the national languages?" — *"German, French and Italian."* Romansh had disappeared.

This is not a flaw specific to Apertus: every large language model writes with the same fluency whether it is right or wrong. On our test set, Apertus asked alone named a wrong or invented article in **59%** of its answers. For a consumer chatbot that is annoying. For a public administration it is disqualifying: a citizen cannot guess that a plausible article number is false, and a disclaimer such as "please verify" only shifts the burden onto them.

The brake on AI in the public sector is therefore neither power nor price. It is trust. And trust cannot be decreed: it has to be verifiable.

A rule everyone understands

Belegt — German for *substantiated, documented* — rests on a single rule: **cite, or stay silent**. Belegt asserts nothing it cannot point to in the official text. Every answer comes with the exact sentences of the Constitution it rests on, highlighted in the article, one click away. If those sentences don't exist, the answer is not given: Belegt says so, explains why, and lists the closest articles for the reader to check — without claiming anything.

"Which article of the Constitution sets the maximum price of bread?" None. An ordinary assistant would invent a plausible article. Belegt answers: *no proof, no answer*, and suggests reading the articles on food security and competition policy.

Who does what

The heart of the project is a clear division of roles. The AI does what it is good at: understanding a question written by a human and rephrasing it. Apertus turns "demonstrate in the street" into "freedom of assembly" in German, French, Italian and Romansh, using the Constitution's own vocabulary. It then drafts a short answer and picks its quotes among the retrieved articles.

Verification is not left to the AI. Simple, deterministic code applies five rules: the cited paragraph exists; the quote appears in it word for word; the quote is long enough to carry meaning; no unproven article number appears in the answer (otherwise the sentence is removed); and if no evidence

survives, Belegt abstains. A sixth check verifies that a search run independently in each of the four languages reaches the same article.

The split is visible on screen: every step carries an “AI” or a “Code” label. Users can see what is probabilistic and what is not.

Four languages, one law

In Switzerland, equality between languages is not a nice-to-have: it is a constitutional requirement. A tool that answers differently depending on the language of the question creates inequality before the law. Belegt treats the four official versions on an equal footing, Romansh included.

We found an elegant way to do it: the texts published by Fedlex carry official identifiers that are identical in every language. Paragraph `art_22/para_1` is the same provision in German, French, Italian and Romansh, so the cross-language check needs no machine translation. Ask in Romansh: the answer comes in Romansh, the quote comes from the official Romansh text, and a panel shows the same provision in the three other languages. If a version diverged, a banner would say so instead of silently choosing. With Belegt, questions asked in all four languages received the same answer in **80%** of cases, against **40%** for Apertus alone.

English questions are welcome too: since English is not an official language of the Confederation, Belegt answers in English and quotes the official French text.

Sovereignty, demonstrated rather than claimed

Everyone says “sovereign”. We wanted to show it. Apertus runs on the machine, with llama.cpp built without network support; the model file is identified by its SHA-256 fingerprint. The Belegt server installs an audit hook that blocks and counts every outbound connection or DNS lookup; the counter is always visible and written into every receipt. Finally, the demo runs inside an isolated Linux network namespace that has no interface except loopback: there is simply no route out. The counter stays at zero, and the answers still arrive.

A receipt for every answer

Every answer produces a readable receipt: which model answered (and its fingerprint), which version of the law was used (and its fingerprint), which evidence was found (each quote with its hash and exact position in the text), which checks passed, and what state the network was in. A legal team or an auditor can download it and file it as supporting evidence. That is what turns a demo into a piece of infrastructure.

Measured, not promised

We don’t say “it works well”. We compared the same local model, on the same questions, with and without Belegt: 81 questions, including 19 traps, a set asked in all four languages, and a held-out set written after our last design change.

	APERTUS ALONE	APERTUS + BELEGT
Cites a correct article	26/62 = 42% [30–54]	**51/62 = 82% [71–90]**
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Abstains although an answer exists	—	9/62 = 15% [8–25]

95% confidence intervals (Wilson). All raw outputs are published in the repository.

These numbers come at a price we accept: Belegt sometimes abstains when an answer existed, because lexical search missed the right article. We prefer one silence too many to one invention.

What Belegt is not, and its limits

Belegt is not a legal adviser: it helps people find and read the official text, not decide an individual case. The verifier guarantees that quotes exist word for word; it does not guarantee that the paraphrase around them is right, nor that the quoted article is the most relevant one. That is why sentences that are not quotes are dotted-underlined, and why a “real quote from the wrong article” counts as an error in our measurements. The model runs at 4 bits on a CPU: an answer takes about 50 seconds (measured median). Our Romansh test questions were written by the team, not by native speakers. And local CPU inference is not bit-for-bit reproducible: in one of our demo recording takes the Romansh question was refused — a wrongful abstention, never an invention.

What Belegt gives back to Apertus

A project that takes from a model without giving anything back is a demo; a project that gives knowledge back is a contribution. Belegt publishes a map of Apertus’s weaknesses (where it cites correctly, where it invents, where it diverges from one language to another), a reusable question set with its traps and its four-language versions, a behaviour worth aiming for in future alignment work — cite or abstain — and a receipt format other projects can adopt. Everything is released under the Apache-2.0 licence.

The Constitution is only a test bed. The method applies to any official text published by Fedlex: the Civil Code, the Data Protection Act, old-age insurance, and tomorrow the municipal regulations a small business or a school has to follow.

Apertus was built open so that trust could be verified. Belegt is a first public demonstration.

Prove it, or stay silent.